

LAB 3**Solve below questions:**

1. Create a superclass Person with attributes name and age, and a method display(). Create a subclass Student that adds an attribute studentID. Write a program to create a Student object and display all its attributes.

PROGRAM:

```
package Lab3;

// Superclass Person
class Person {
    // Person class attributes
    String name = "Ashi";
    int age = 20;

    // Method to display the attributes in Person class
    public void display() {
        System.out.println("The name of the person is " + name + " and the
age is " + age);
    }
}

// Subclass Student
public class Student extends Person {

    // Additional attribute of Student class
    int studentID = 1;

    public static void main(String[] args) {
        // Student class object
        Student s1 = new Student();
        // Printing attributes
        System.out.println("Student details: ");
        System.out.println("Name : " + s1.name);
        System.out.println("Age : " + s1.age);
        System.out.println("StudentID : " + s1.studentID);
        s1.display();
    }
}
```

OUTPUT:

```
<terminated> Student (1) [Java Application] C:\Users\Ashi chouhan\p2\pool\
Student details:
Name : Ashi
Age : 20
StudentID : 1
The name of the person is Ashi and the age is 20
```

2. Create a superclass Calculator with a method add(int a, int b). Create a subclass AdvancedCalculator that overloads the add method to handle three integers.

PROGRAM:

```
package Lab3;

// Superclass calculator
class Calculator {

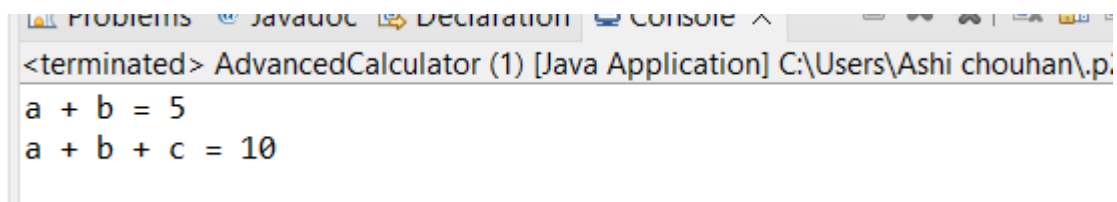
    // Method to add 2 integers a and b
    public int add(int a, int b) {
        return a+b;
    }
}

// Subclass of Calculator class
public class AdvancedCalculator extends Calculator {

    // Overriding add(int a, int b) method from superclass to add 3 integers
    public int add(int a, int b, int c) {
        return a+b+c ;
    }

    public static void main(String[] args) {
        int a = 2;
        int b = 3;
        int c = 5;
        // object Of AdvancedCalculator class
        AdvancedCalculator ad = new AdvancedCalculator();
        // Printing output
        System.out.println("a + b = " + ad.add(a, b));
        System.out.println("a + b + c = " + ad.add(a, b, c));
    }
}
```

OUTPUT:



3. Create a superclass Vehicle with a method move(). Create subclasses Car and Bike that inherit from Vehicle. Write a program to create objects of Car and Bike and call the move() method on each.

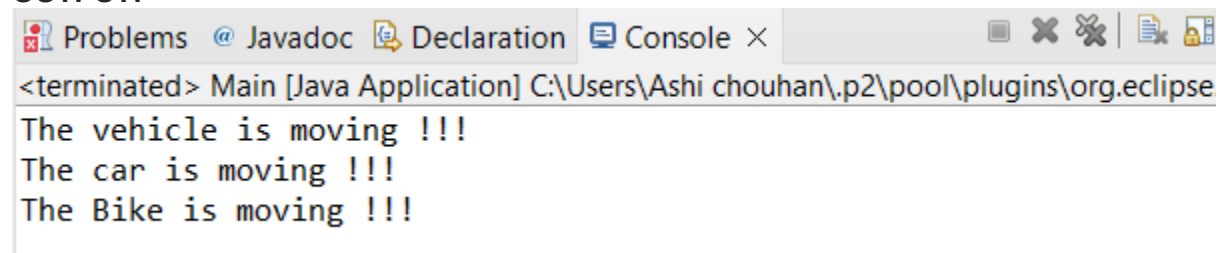
PROGRAM:

```
package nDemo;

// Superclass Vehicle
class Vehicle {
    public void move() {
        System.out.println("The vehicle is moving !!!");
    }
}
```

```
    }  
}  
  
// Subclass of Vehicle class  
class Car extends Vehicle {  
  
    public void move() {  
        System.out.println("The car is moving !!!");  
    }  
}  
  
//Subclass of Vehicle class  
class Bike extends Vehicle {  
  
    public void move() {  
        System.out.println("The Bike is moving !!!");  
    }  
}  
  
public class Main {  
  
    public static void main(String[] args) {  
  
        // Object of Vehicle class  
        Vehicle v1 = new Vehicle();  
        // Object of Car class  
        Car c1 = new Car();  
        // Object of Bike class  
        Bike b1 = new Bike();  
  
        // Calling move() method on each  
        v1.move();  
        c1.move();  
        b1.move();  
    }  
}
```

OUTPUT:



The screenshot shows the Eclipse IDE's Console window. The title bar includes tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console output is as follows:

```
<terminated> Main [Java Application] C:\Users\Ashi chouhan\.p2\pool\plugins\org.eclipse  
The vehicle is moving !!!  
The car is moving !!!  
The Bike is moving !!!
```

4. Create an class Employee with an abstract method calculatePay(). Create subclasses SalariedEmployee and HourlyEmployee that implement the calculatePay() method. Write a program to create objects of both subclasses and call the calculatePay() method.

PROGRAM:

```
package Lab3;  
  
//Abstract class Employee  
abstract class Employee {
```

```
        public abstract double calculatePay();
    }

    //Subclass of Employee
    class SalarizedEmployee extends Employee {
        private double salary;

        // Parameterized constructor
        public SalarizedEmployee(double salary) {
            this.salary = salary;
        }

        public double calculatePay() {
            return salary;
        }
    }

    // Subclass of Employee
    class HourlyEmployee extends Employee {
        private double hourlyRate;
        private int hours;

        // Constructor
        public HourlyEmployee(double hourlyRate, int hoursWorked) {
            this.hourlyRate = hourlyRate;
            this.hours = hoursWorked;
        }

        public double calculatePay() {
            return hourlyRate * hours;
        }
    }

    public class EmployeePay {

        public static void main(String[] args) {
            // Creating objects of SalarizedEmployee and HourlyEmployee
            SalarizedEmployee sE = new SalarizedEmployee(5000);
            HourlyEmployee hE = new HourlyEmployee(25.0, 160);

            // To call calculatePay by both subclass object
            double salariedPay = sE.calculatePay();
            System.out.println("Salaried Employee Pay: Rs " + salariedPay);

            double hourlyPay = hE.calculatePay();
            System.out.println("Hourly Employee Pay: Rs " + hourlyPay);
        }
    }
}
```

OUTPUT:

```

C:\Users\Ashi Chouhan\Desktop> java EmployeePayApp
Salaried Employee Pay: Rs 5000.0
Hourly Employee Pay: Rs 4000.0

```

5. Create an class Document with an method void open(). Implement subclasses WordDocument, PDFDocument, and SpreadsheetDocument that extend Document and provide implementations for open(). Write a main class to demonstrate opening different types of documents.(implement compile time- polymorphism).

PROGRAM:

```

package Lab3;

// Superclass Document
class Document {
    public void open() {
        System.out.println("Opening Document !!!");
    }
}

// Subclass of Document class
class WordDocument extends Document {
    public void open() {
        System.out.println("Opening Word Document !!!");
    }
}

//Subclass of Document class
class PDFDocument extends Document {
    public void open() {
        System.out.println("Opening PDF Document !!!");
    }
}

//Subclass of Document class
class SpreadsheetDocument extends Document {
    public void open() {
        System.out.println("Opening Spreadsheet Document !!!");
    }
}

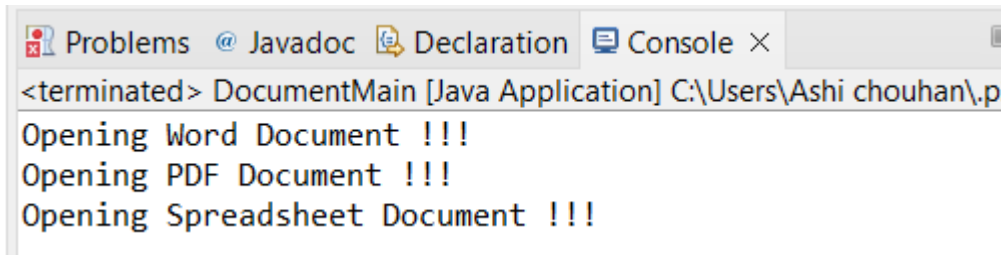
public class DocumentMain {

    public static void main(String[] args) {
        // Creating objects of all subclasses
        WordDocument WD = new WordDocument();
        PDFDocument PDF = new PDFDocument();
        SpreadsheetDocument SD = new SpreadsheetDocument();

        // Calling open() method using all objects
        WD.open();
        PDF.open();
        SD.open();
    }
}

```

```
}
```

OUTPUT:


```
<terminated> DocumentMain [Java Application] C:\Users\Ashi chouhan\.p
Opening Word Document !!!
Opening PDF Document !!!
Opening Spreadsheet Document !!!
```

6. Create a class Calculator with overloaded methods add() that take different numbers and types of parameters: int add(int a, int b), double add(double a, double b), int add(int a, int b, int c) Write a main class to demonstrate the usage of these methods.

PROGRAM:

```
package Lab3;

class Calculator2 {
    // Method to add two integers
    public int add(int a, int b) {
        return a + b;
    }

    // Method to add two doubles
    public double add(double a, double b) {
        return a + b;
    }

    // Method to add three integers
    public int add(int a, int b, int c) {
        return a + b + c;
    }
}

public class CalculatorMain {
    public static void main(String[] args) {
        // Object of Calculator2 class
        Calculator2 c1 = new Calculator2();

        // Function call and printing outputs
        System.out.println("2 + 3 = " + c1.add(2, 3));
        System.out.println("2.13 + 5.67 = " + c1.add(2.13, 5.67));
        System.out.println("2 + 3 + 5 = " + c1.add(2, 3, 5));
    }
}
```

OUTPUT:

```
<terminated> CalculatorMain [Java Application] C:\Users\Ashi chouhan\p2\
2 + 3 = 5
2.13 + 5.67 = 7.8
2 + 3 + 5 = 10
```

7. Create a [JavaBean](#) class Person with properties firstName, lastName, age, and email. Implement the required no-argument constructor, getter and setter methods for each property. Write a main class to create an instance of Person, set its properties, and print them out.

OUTPUT:

```
package Lab3;

import java.io.Serializable;

class Person1 implements Serializable {
    // Private attribute declaration
    private String firstName;
    private String lastName;
    private int age;
    private String email;

    Person1() {
        // Default constructor
    }

    // Getter setter Method for firstName
    public void setFName(String firstName) {
        this.firstName = firstName;
    }

    public String getFName() {
        return firstName;
    }

    // Getter setter Method for lastName
    public void setLName(String lastName) {
        this.lastName = lastName;
    }

    public String getLName() {
        return lastName;
    }

    // Getter setter Method for age
    public void setAge(int age) {
        this.age = age;
    }

    public int getAge() {
        return age;
    }
}
```

```
// Getter setter Method for email
public void setEmail(String email) {
    this.email = email;
}

public String getEmail() {
    return email;
}
}

public class PersonMain {

    public static void main(String[] args) {
        // Object of Person1 class
        Person1 p1 = new Person1();
        // Setting values of all attributes
        p1.setFName("Ashi");
        p1.setLName("Chouhan");
        p1.setAge(20);
        p1.setEmail("chouhancrusador@gmail.com");

        // Retrieving or getting all attributes
        System.out.println("First Name : " + p1.getFName());
        System.out.println("Last Name : " + p1.getLName());
        System.out.println("Age : " + p1.getAge());
        System.out.println("Email : " + p1.getEmail());

    }

}
```

OUTPUT:

```
<terminated> PersonMain [Java Application] C:\Users\Ashi chouhan\.p2\pc
First Name : Ashi
Last Name : Chouhan
Age : 20
Email : chouhancrusador@gmail.com
```

8. Create a [JavaBean](#) class Car with properties make, model, year, and color. Implement the required no-argument constructor, getter and setter methods for each property. Write a main class to create an instance of Car, set its properties, and print the car details.

PROGRAM:

```
package Lab3;

class Car1 {
    private String make;
    private String model;
```

```
private int year;
private String color;

// No-argument constructor
public Car1() {

}

// Getter and Setter methods for 'make'
public String getMake() {
    return make;
}

public void setMake(String make) {
    this.make = make;
}

// Getter and Setter methods for 'model'
public String getModel() {
    return model;
}

public void setModel(String model) {
    this.model = model;
}

// Getter and Setter methods for 'year'
public int getYear() {
    return year;
}

public void setYear(int year) {
    this.year = year;
}

// Getter and Setter methods for 'color'
public String getColor() {
    return color;
}

public void setColor(String color) {
    this.color = color;
}

}

public class CarMain {
    public static void main(String[] args) {
        // Object of Car1 class
        Car1 c1 = new Car1();

        // Set properties using setter methods
        c1.setMake("Toyota");
        c1.setModel("Camry");
        c1.setYear(2022);
        c1.setColor("Silver");

        // Getting or retrieving all properties
        System.out.println("\nIndividual Properties:");
    }
}
```

```
        System.out.println("Make: " + c1.getMake());  
        System.out.println("Model: " + c1.getModel());  
        System.out.println("Year: " + c1.getYear());  
        System.out.println("Color: " + c1.getColor());  
    }  
}
```

OUTPUT:

```
<terminated> CarMain [Java Application] C:\Users\Ashi chouhan\.p2\pool\plu  
Individual Properties:  
Make: Toyota  
Model: Camry  
Year: 2022  
Color: Silver
```